

Course code: ECA3102

Course: Nano electronics

Exp 1: Simulation of Density of States of Quantum well structure with various nano meter thickness.

Why ?

The purpose of this project is to understand how quantum confinement changes the Density of States (DOS) when the thickness of a quantum well is varied from a few nanometers to several nanometers. The Density of States tells us How many electron energy states are available at each energy level.

Basic Model Configuration

Advanced Model Configuration

Temperature: ☐ 300K

Temperature Gradient: ☐ 10K

Silicon Doping: $1e+18/cm^3$

Germanium Doping: $1e+18/cm^3$

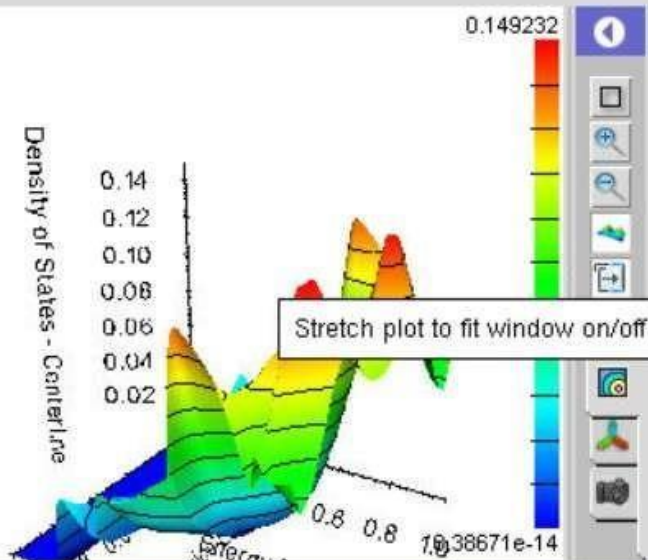
Substrate Composition $Si(x)Ge(1-x)$: 1

Mat 1 Material: Silicon

Mat 2 Material: Germanium

Simulate

Result: Density of States - Centerline



2 results

Clear One

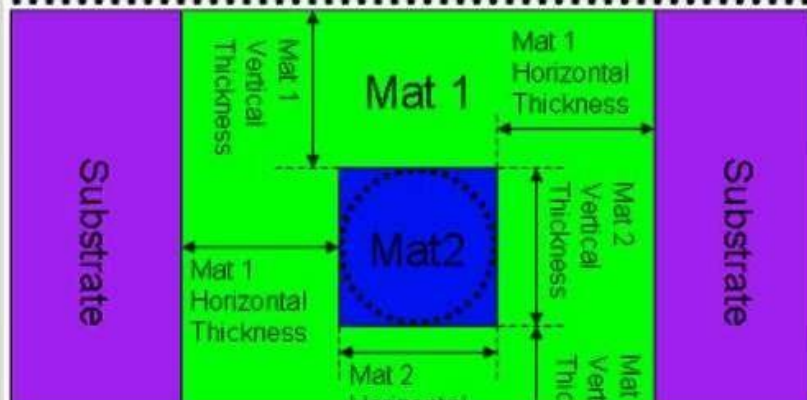
Clear All

All

Simulation = #2

Mat 1 Horizontal Thickness = 0.2nm
Mat 1 Vertical Thickness = 0.2nm
Mat 2 Horizontal Thickness = 0.2nm
Mat 2 Vertical Thickness = 0.2nm
barrierLx = 0.2nm
barrierLy = 0.2nm
wellLx = 0.2nm
wellLy = 0.2nm

Reflective Boundary Condition



Storage (manage)

0% of 10.000004GB



780 × 600

Case study 1: Simulate the density of states of quantum well of thickness 0.5 nm

Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.5 nm well thickness** and analyze the effect of strong quantum confinement on the available electron energy states.

Thermoelectric Power Factor Calculator
for Nanocrystalline Composites

✕ Terminate ➡ Keep for later

Basic Model Configuration Advanced Model Configuration

Temperature: 300K

Temperature Gradient: 10K

Silicon Doping: 1e+18/cm3

Germanium Doping: 1e+18/cm3

Substrate Composition Si(x)Ge(1-x): 1

Mat 1 Material: Silicon

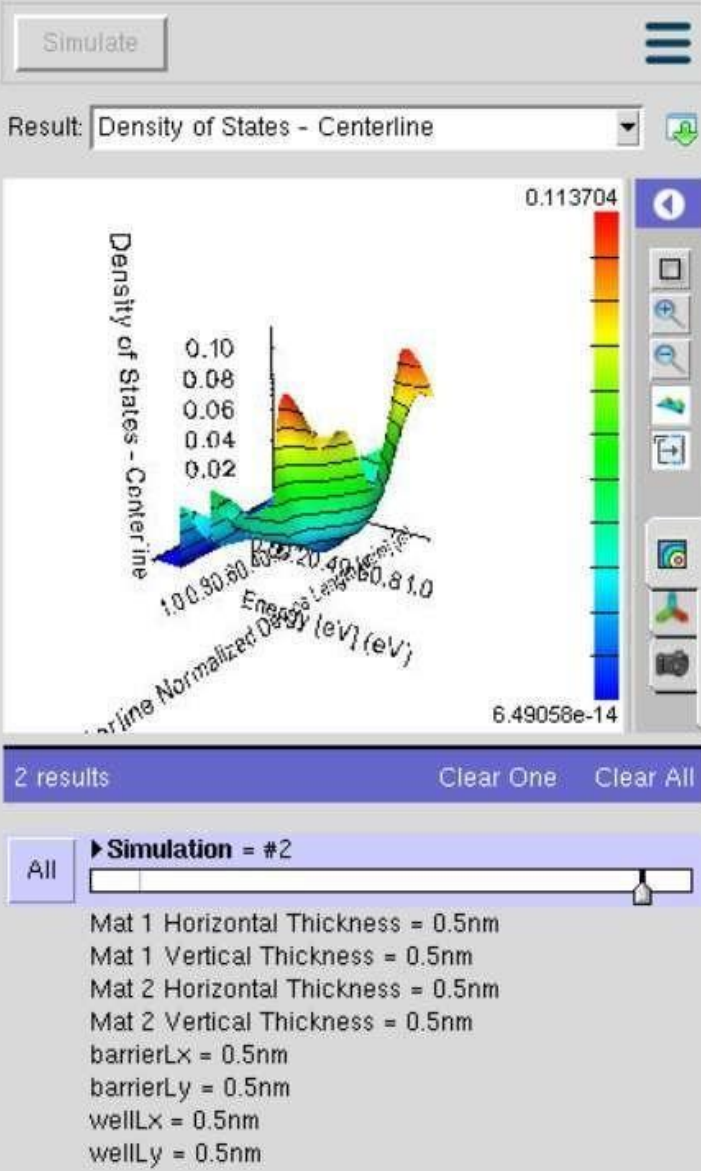
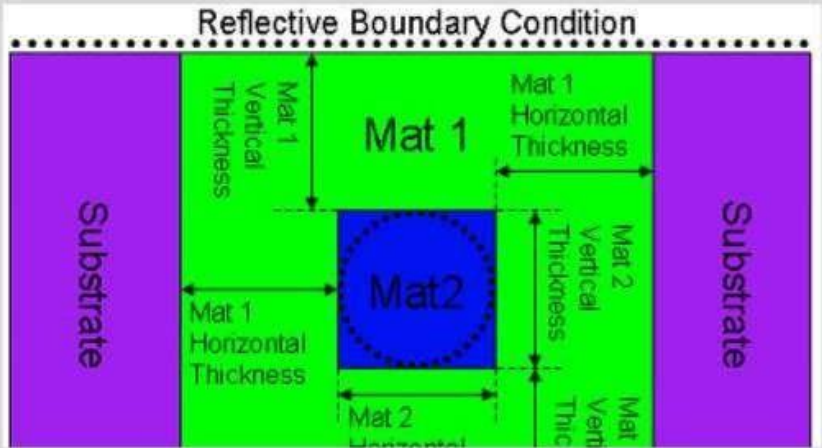
Mat 2 Material: Germanium

Mat 1 Horizontal Thickness: 0.5nm

Mat 1 Vertical Thickness: 0.5nm

Mat 2 Horizontal Thickness: 0.5nm

Mat 2 Vertical Thickness: 0.5nm



Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm

Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.6 nm well thickness** and investigate the effect of increased well thickness on the electronic energy states.

for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature: ☐ 300K

Temperature Gradient: ☐ 10K

Silicon Doping:

Germanium Doping:

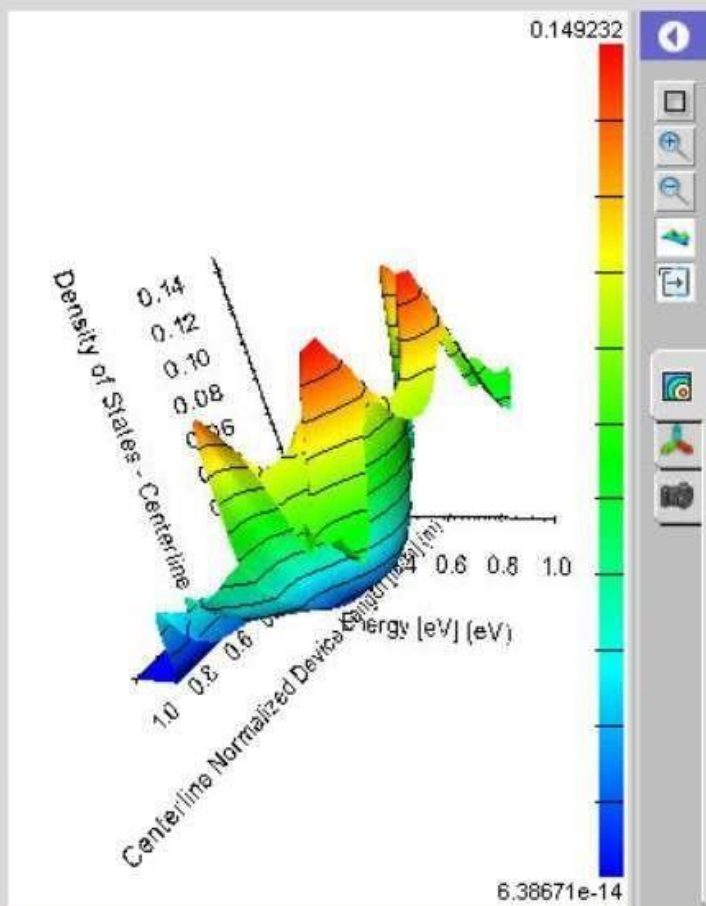
Substrate Composition Si(x)Ge(1-x):

Mat 1 Material:

Mat 2 Material:

Simulate

Result:



1 result

Clear

Storage (manage)

0% of 10.000004GB



780 × 600

The simulation of the **Density of States (DOS)** for quantum well structures with different nanometer thicknesses demonstrates that **quantum confinement strongly influences the electronic properties** of semiconductor nanostructures. As the well thickness decreases, the energy levels become more discrete and widely spaced, resulting in significant changes in the Density of States. Conversely, increasing the well thickness reduces the confinement effect, causing the DOS to gradually approach that of a bulk semiconductor. These results confirm that **quantum well thickness is a critical design parameter** for tailoring the electronic and optical characteristics of devices such as quantum well lasers, LEDs, photodetectors, and other nanoelectronic applications.